Algorithmic Explanation of Phenomenal Consciousness: A Limitation Result and Its Most Parsimonious Interpretation

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Abstract

This paper develops a formal framework for evaluating the explanatory completeness of theories of consciousness and argues that the principled thesis—that phenomenal consciousness is not completely algorithmically capturable—is the most parsimonious interpretation of the current state of the field. Three levels of explanatory success are distinguished: weak capture, structural capture, and strong capture. Existing theories, including Global Workspace Theory, Integrated Information Theory, and Recurrent Processing Theory, are shown to achieve at most structural capture while falling short of explanatory closure. The framework identifies three independently motivated constraints—representability, algorithmic realizability, and explanatory closure—whose simultaneous satisfaction is required for complete strong capture. Each has been independently challenged from different disciplines, and none has been met. The persistent, independent, and multi-dimensional character of this triple failure is more parsimoniously explained by a principled limitation than by three coincidental gaps. The conclusion is not that this is proved, but that it is better supported than its negation given the current evidence; the framework specifies the precise conditions whose satisfaction would overturn it. If the principled thesis holds, the conditional limitation becomes unconditional: the algorithmic framework within which human explanation operates cannot completely explain phenomenal consciousness. The result is situated within a layered analysis of realization, simulation, instantiation, and attribution, and engages objections from type-A and type-B physicalism, intertheoretic reduction, illusionism, and mathematical consciousness science.

Keywords: phenomenal consciousness, hard problem, explanatory gap, algorithmic explanation, computational theory of mind, artificial consciousness, philosophy of mind, inference to the best explanation

1 Introduction

The hard problem of consciousness concerns the relation between physical processes and subjective experience Chalmers (1995), and the associated explanatory gap between them Levine (1983). While empirical research has made substantial progress in identifying neural correlates, it remains unclear whether such progress contributes to explaining why experience exists at all.

This paper makes four contributions. First, it develops a framework of *levels of capture*—weak, structural, and strong—for evaluating what theories of consciousness achieve in terms of explanatory completeness. Second, it identifies three independently motivated constraints whose simultaneous satisfaction is required for complete strong algorithmic capture. Third, it argues that the persistent, independent failure of all three constraints is more parsimoniously explained by the principled thesis that phenomenal consciousness is not algorithmically capturable than by the hypothesis that three independent gaps will all eventually close—and that if this principled thesis holds, the limitation on algorithmic explanation becomes unconditional. Fourth, it distinguishes levels of artificial realization and connects these to the explanatory framework, showing how attributions of consciousness to artificial systems can systematically outrun the evidential basis for instantiation.

The paper's central contribution is the third: it takes a position, albeit a qualified one. The framework specifies exactly what would need to be established to refute the limitation—representability, algorithmic realizability, and explanatory closure must all be demonstrated simultaneously. The current state of the field, evaluated against these conditions, favors the principled thesis over its negation. This claim is inductive, not deductive. It may be overturned by future work that satisfies the specified conditions, but it is a claim about what the evidence currently supports, not merely a conditional exercise.

The formal vocabulary is meta-theoretic: it does not compete with object-level theories such as Integrated Information Theory or Global Workspace Theory, but provides a framework for evaluating what any such theory achieves. The formal result is stated in Section 4 before the case for its key premise is developed in Section 5, so that the reader can assess what the framework requires before evaluating whether those requirements are met. The analysis is addressed to those who hold that phenomenal consciousness poses a genuine explanatory problem; for those who deny this (Section 8.1), the framework is inapplicable rather than refuted.

2 Preliminaries

2.1 Phenomenal and Access Consciousness

Following Block (1995), consciousness divides into at least two conceptually distinct notions: phenomenal consciousness (subjective experience, or what it is like to be in a state) and access consciousness (availability of information for reasoning, report, and control of behavior). Block argues that these are conceptually independent, motivating the possibility of dissociation: states that are phenomenally conscious but not access-conscious, or vice versa.

This distinction is not merely terminological. Block's overflow argument (2011) suggests that phenomenal consciousness may exceed cognitive access. Empirical paradigms such as Sperling's partial report task indicate that subjects may have richer perceptual experience than can be reported or maintained in working memory. However, this claim remains debated: Yang (2025) argues that evidence interpreted as supporting fragmentation may be consistent with rich phenomenology, and Overgaard (2018), Naccache (2018), and Cohen and Dennett (2011) contend that phenomenal consciousness reduces to access.

Neuroscientific accounts reinforce the possibility of dissociation. Recurrent processing theory Lamme (2018) proposes distinct neural mechanisms for phenomenal and access consciousness: early recurrent processing in sensory areas may support phenomenal experience, while later global ignition supports access. This suggests a possible separation at the level of neural realization.

Throughout, C denotes phenomenal consciousness: the aspect of experience that is not exhausted by functional accessibility. This restriction aligns the analysis with the explanatory gap and the hard problem rather than with cognitive or behavioral capacities alone. It is a scope condition: the claims that follow are addressed to phenomenal consciousness specifically. If phenomenal consciousness, so defined, does not exist, the framework loses its target rather than being refuted (Section 8.6).

2.2 Epistemic, Representational, and Computational Constraints

Any theory of consciousness faces an asymmetry of access. Conscious experience is available through first-person introspection; mechanisms and correlates are available through third-person observation and experiment. Neither mode, taken alone, provides unrestricted access to the full body of truths about consciousness Nagel (1974); Chalmers (1995); Wu and Bhatt (2025); Seth and Bayne (2022).

Epistemic limitations (limits on what can be known or observed) must be distinguished from computational limitations (limits on what can be formally derived or algorithmically captured). An epistemic gap does not automatically imply a computational one, and vice versa Cleland (1993); Copeland (2024); Wagner-Altendorf (2024). The framework developed here concerns both, but gives special attention to cases in which they converge.

A further constraint concerns representability. Let $\mathcal{R}$ denote the class of structures that admit formal representation, and let $\mathcal{A}$ denote the class of theories whose explanatory relations are Turing-computable. Since anything algorithmically capturable must first be representable ($\mathcal{A} \subseteq \mathcal{R}$), a failure of representability entails a failure of algorithmic capture. This becomes critical when considering whether the full truth-set of consciousness can be brought within the scope of a formal theory Chalmers (1996); McGinn (1989); Gu (2025).

Definition of algorithmic theories. The notion of algorithmic theory used here is intentionally weaker than full formal axiomatizability. A theory $T \in \mathcal{A}$ if the mappings, derivations, or procedures by which T relates explanans to explanandum can be implemented by a Turing machine. The requirement is not that a theory be expressible within a fixed formal system, but that its explanatory content be effectively computable. This aligns the notion of algorithmic explanation with the standard interpretation of effective procedure in computability theory Turing (1936); Copeland (2024).

2.3 The Empirical-Conceptual Distinction

The explanatory problem of consciousness contains at least two distinct targets: an empirical target (identifying mechanisms, correlates, and functional organization) and a conceptual target (explaining why and how phenomenal experience exists at all). These should not be conflated Chalmers (1995, 1996); Wagner-Altendorf (2024). Identifying neural correlates does not by itself settle the question of whether phenomenal consciousness has been explained.

Empirical work suggests that consciousness is structured and distributed. Phenomenal consciousness may therefore be treated, at least schematically, as internally differentiated:

$$C = (c_1, c_2, \dots, c_n)$$

where the c_i denote distinguishable components, dimensions, or aspects of conscious content. Let

$$\phi : C \to \mathcal{N}$$

map conscious contents to their neural correlates or realizations. At minimum, distinct conscious contents need not collapse onto a single neural state:

$$\phi(c_i) \neq \phi(c_j) \quad \text{for at least some } i \neq j$$

In particular, there need not exist a single neural item $N \in \mathcal{N}$ whose inverse image under ϕ recovers the whole structured field of conscious content:

$$\nexists N \in \mathcal{N} \text{ such that } \phi^{-1}(N) = C$$

This is a schematic way of expressing distributed realization, not a claim that current neuroscience has established a mathematically exact map of this form. The point is that local neural correlates do not by themselves amount to a complete capture of the phenomenal field Wu and Bhatt (2025); Doerig et al. (2024); Cogitate Consortium et al. (2025). This empirical picture supports the possibility that mechanistic descriptions capture part but not all of what needs to be explained.

No single explanatory framework has been shown to exhaust the full truth-set of consciousness. Compositional frameworks may explain relations among parts, contents, and mechanisms while failing to explain why organized structures are accompanied by experience. Instantiational frameworks may specify conditions for consciousness while failing to explain why those conditions yield phenomenal consciousness rather than mere functional organization Searle (1992); Dung (2025); Multiple Authors (2025). This pattern of complementary failure recurs across the field.

3 A Framework for Explanatory Capture

The central claim of this paper concerns the limits of explaining consciousness. To state that claim precisely, it is necessary to distinguish different strengths of explanatory success.

3.1 Truths About Consciousness

Let Truth(C) denote the totality of non-explanatory truths about phenomenal consciousness. At minimum, this includes truths about the occurrence of phenomenal states, truths about the internal structure of conscious content, and truths about relations between phenomenal states and their neural or functional correlates.

Let Func(C) denote the functional or access-related description of consciousness: the role a state plays in reasoning, report, control, and information availability. Functional truths form part of the broader body of truths about consciousness:

$$\text{Func}(C) \subsetneq \text{Truth}(C)$$

Whether functional description exhausts Truth(C) depends on whether phenomenal consciousness is reducible to access or functional organization. If phenomenal consciousness is not so reducible—as the distinction in Section 2.1 motivates—then:

$$\mathrm{Func}(C) \neq \mathrm{Truth}(C)$$

This distinction is important because progress in identifying neural correlates and functional mechanisms constitutes progress on Func(C), but does not by itself settle the question of whether the full truth-set Truth(C) has been captured:

$$\mathrm{Progress}(\mathrm{Func}(C)) \nRightarrow \mathrm{Progress}(\mathrm{Truth}(C))$$

We distinguish these from a further class of potential truths, $\mathrm{Truth}_{\mathrm{why}}(C)$, which concern why phenomenal consciousness accompanies certain physical or functional states at all. The status of these truths is philosophically contested. The requirement of explanatory closure presupposes that such truths are non-empty. If no such truths exist, the limitation result holds trivially, because the explanatory demand has no corresponding target.

The specification of Truth(C) requires clarification. The framework does not require that all truths about consciousness be enumerable in advance. It requires only that, conditional on phenomenal realism, there *are* facts of the matter about phenomenal consciousness. The truth-set is defined extensionally: whatever the facts about phenomenal experience turn out to be, they constitute Truth(C). This is no more circular than the requirement that a complete theory of quantum gravity cover all truths about quantum gravity, even though those truths cannot currently be enumerated. The negative result does not depend on knowing *which* specific truths are in Truth(C); it depends on the structural observation that no existing theory simultaneously achieves the conditions required for complete capture of *whatever* those truths turn out to be.

3.2 Levels of Capture

Three levels of capture are distinguished.

Weak capture. A theory T weakly captures consciousness if it reliably tracks, predicts, or correlates with conscious states:

$$\mathrm{Capture}_{\mathrm{w}}(T, C)$$

This includes the identification of neural correlates, behavioral signatures, or functional markers.

Structural capture. A theory T structurally captures consciousness if it reproduces or preserves significant aspects of the internal organization of conscious content:

$$\mathrm{Capture}_{\mathrm{s}}(T, C)$$

A structurally capturing theory does not merely detect consciousness, but models relevant relations among its contents, dimensions, or transitions.

Strong capture. A theory T strongly captures consciousness if it achieves explanatory closure *and* its scope includes the relevant truths about consciousness:

$$\mathrm{Capture}_{\mathrm{str}}(T, C) \iff \mathrm{ExplClosure}(T, C) \wedge \mathrm{Truth}(C) \subseteq \mathrm{Scope}(T)$$

where ExplClosure(T, C) means that T explains why phenomenal consciousness exists at all, not merely that it correlates with or models its structure.

3.3 The Capture Hierarchy

These notions stand in a hierarchical relation:

$$\mathrm{Capture}_{\mathrm{str}}(T, C) \Rightarrow \mathrm{Capture}_{\mathrm{s}}(T, C) \Rightarrow \mathrm{Capture}_{\mathrm{w}}(T, C)$$

but not conversely. Reliable correlation does not explain why experience exists, and structural preservation does not yield phenomenal explanatory closure.

The implication

$$\mathrm{Capture}_{\mathrm{str}}(T, C) \Rightarrow \mathrm{Capture}_{\mathrm{s}}(T, C)$$

warrants brief justification. Strong capture requires truth-completeness: $\mathrm{Truth}(C) \subseteq \mathrm{Scope}(T)$. Since $\mathrm{Truth}(C)$ includes truths about the internal structure of conscious content—spatial organization of the visual field, qualitative differences among modalities, relations among phenomenal states—any theory whose scope includes these truths must preserve significant aspects of that structure. A theory that explains why consciousness exists without modeling its structure (for example, by positing it as a fundamental property) may achieve explanatory closure but not truth-completeness, and therefore does not achieve strong capture.

A theory T completely explains consciousness only if it achieves $\mathrm{Capture}_{\mathrm{str}}(T, C)$. This makes explicit that the paper does not deny the possibility of weak or structural theories. The target is the stronger claim that consciousness can be completely explained in the sense relevant to the hard problem.

3.4 Why Explanatory Closure is Structurally Resistant

Physical and functional descriptions operate in a third-person, structural vocabulary: they specify relations, mechanisms, and causal organization. Phenomenal consciousness, by contrast, is essentially first-personal and perspectival. Let $\mathcal{D}_{\mathrm{struct}}$ and $\mathcal{D}_{\mathrm{phen}}$ denote the domains of structural and phenomenal description, respectively. Explanatory closure requires a mapping $f : \mathcal{D}_{\mathrm{struct}} \to \mathcal{D}_{\mathrm{phen}}$ that not only correlates structures with experiences, but explains why the former are accompanied by the latter Levine (1983).

Phenomenal contents appear to involve irreducibly perspectival or indexical elements that are not straightforwardly eliminable in favor of third-person structural descriptions without loss of what is to be explained Nagel (1974); Levine (1983); Chalmers (1996). Algorithmic theories operate over formally representable and intersubjectively accessible structures. Thus, explanatory closure requires bridging a domain that is structurally characterizable with one that is essentially perspectival. No generally accepted method eliminates this gap without either reducing phenomenal content to structure or leaving the explanatory demand unmet.

Even if this gap is ultimately cognitive rather than metaphysical, the requirement of explanatory closure remains unsatisfied at the level of available theories. This suffices, within the present framework, to block complete strong capture. A type-A physicalist will deny that there is a genuine explanatory demand here; this objection is addressed in Section 8.1.

3.5 Worked Examples

To illustrate the capture hierarchy, consider three leading theories of consciousness.

Global Workspace Theory. GWT Baars (1988); Dehaene (2014) identifies global broadcasting of information across specialized processors as the basis of conscious access. It reliably tracks which stimuli reach conscious access (weak capture) and preserves aspects of the internal organization of conscious content, including the transition from local processing to global availability and the serial character of access (structural capture). However, GWT does not explain why global broadcasting is accompanied by phenomenal experience rather than occurring without any subjective character. A system could, in principle, broadcast information globally while remaining without phenomenal experience. Thus $\mathrm{Capture}_{\mathrm{s}}(T_{GWT}, C)$ but $\neg\,\mathrm{ExplClosure}(T_{GWT}, C)$.

A defender of GWT may respond that broadcasting *constitutes* consciousness. Within the present framework, this amounts to structural capture unless it explains why this structure is accompanied by phenomenal experience. The demand is not for an additional mechanism, but for an account of the constitutive relation itself.

Integrated Information Theory. IIT Tononi et al. (2016); Albantakis et al. (2023) specifies a quantity (Φ) associated with consciousness and characterizes the structural conditions under which integrated information arises. It achieves structural capture: it models formal relations among elements of a system and identifies conditions under which those relations yield high Φ. But IIT does not explain why systems with high Φ are accompanied by phenomenal experience. It postulates that integrated information *is* consciousness, but the explanatory step from a mathematical quantity to phenomenal experience remains stipulative rather than derived. Thus $\mathrm{Capture}_{\mathrm{s}}(T_{IIT}, C)$ but $\neg\,\mathrm{ExplClosure}(T_{IIT}, C)$.

Recurrent Processing Theory. RPT Lamme (2018) identifies local recurrent activity in sensory cortex as the neural basis of phenomenal consciousness. It achieves weak and structural capture: it tracks a specific neural mechanism and models the processing dynamics associated with phenomenal states. But the question of why recurrent processing gives rise to experience rather than merely to enhanced information processing remains open. Thus $\mathrm{Capture}_{\mathrm{s}}(T_{RPT}, C)$ but $\neg$ $\mathrm{ExplClosure}(T_{RPT}, C)$.

The pattern extends beyond these three theories. Higher-order theories Rosenthal (2005); Brown et al. (2019), which hold that consciousness consists in a higher-order representation of a first-order mental state, achieve structural capture of the representational organization of consciousness but do not explain why having a higher-order representation of a state should be accompanied by phenomenal experience rather than being merely a computational relation among representations.

In each case, the theory achieves weak and structural capture while falling short of explanatory closure. This is not an accident or an artifact of incomplete empirical work. It reflects a recurring structural pattern: the explanatory gap persists precisely where the transition from mechanism to experience is required.

4 The Main Result

4.1 Formal Framework

Let $\mathcal{A}$ denote algorithmic theories, $\mathcal{E}$ denote admissible explanations, and C denote phenomenal consciousness as defined above.

$$\begin{aligned} A1 &: \mathcal{E} \subseteq \mathcal{A} \\ A2' &: \neg\exists T \in \mathcal{A} \text{ such that } \mathrm{Capture}_{\mathrm{str}}(T, C) \\ A3 &: \text{A theory completely explains } C \text{ only if } \mathrm{Capture}_{\mathrm{str}}(T, C) \end{aligned}$$

4.2 Result

Theorem 1. *If A1, A2′, and A3 hold, then no admissible explanation completely explains phenomenal consciousness in the sense of complete strong capture.*

Proof. Suppose for contradiction that some $T \in \mathcal{E}$ completely explains C. By A3, $\mathrm{Capture}_{\mathrm{str}}(T, C)$. By A1, $T \in \mathcal{A}$. This contradicts A2′. □

The deductive step from these assumptions to the conclusion is immediate. The substantive contribution lies not in the derivation but in the conceptual framework that gives the assumptions content, in the inductive case for A2′, and in the argument that the evidence favors the stronger principled thesis A2 over its negation.

4.3 On the Role of A1

Assumption A1 is not required for the core observation that no $T \in \mathcal{A}$ achieves complete strong capture—this follows from A2′ alone. A1's role is to extend the result from algorithmic theories to the space of admissible explanation: if all explanations are algorithmic, then the limitation applies not merely to a class of theories but to explanation itself. This extension is substantive only if one allows for the possibility of non-algorithmic explanations; otherwise A1 is an identity. The algorithmic status of explanation is discussed further in Section 6.

4.4 What the Result Does and Does Not Show

The result does not rule out weak or structural capture, predictive success, empirical modeling, or even some forms of structural reproduction or instantiation. What it targets, conditionally, is complete strong algorithmic explanation—the conjunction of explanatory closure and truth-completeness within an algorithmic framework.

The result is conditional on all three assumptions. Of these, A2′ carries the paper's substantive weight. The following section develops the case for A2′ and argues that the evidence points toward the stronger thesis A2.

5 The Case for A2′ and the Direction Toward A2

The operational premise A2′ asserts that no existing algorithmic theory has achieved complete strong capture of phenomenal consciousness. The stronger thesis $A2 : \mathrm{Truth}(C) \notin \mathcal{A}$ asserts that complete strong algorithmic capture is impossible in principle. This section motivates A2′, examines what the persistent truth of A2′ indicates, and argues that A2 is the most parsimonious interpretation of the available evidence.

5.1 Three Necessary Conditions

For an algorithmic theory to achieve complete strong capture, at least three conditions must hold simultaneously:

(a) **Representability:** $\mathrm{Truth}(C) \subseteq \mathcal{R}$—the truths about phenomenal consciousness must admit formal representation.

(b) **Algorithmic realizability:** the explanatory relations connecting phenomenal and physical or functional descriptions must be algorithmically derivable.

(c) **Explanatory closure:** the resulting theory must explain why phenomenal consciousness exists, not merely describe its structure.

These correspond to the formal availability of the explanandum, the computational realizability of explanatory relations, and the satisfaction of the explanatory demand characteristic of the hard problem. Failure of any one is sufficient to block complete strong capture.

Proposition 2 (Triple Constraint)**.** *For any theory $T \in \mathcal{A}$ to achieve* $\mathrm{Capture}_{\mathrm{str}}(T, C)$*, all three of the following must hold simultaneously: (a)* $\mathrm{Truth}(C) \subseteq \mathcal{R}$*, (b) the explanatory relations of T are algorithmically derivable, and (c) T achieves* $\mathrm{ExplClosure}(T, C)$*. Failure of any one condition is sufficient to block complete strong capture.*

This decomposition is analytic—it unpacks what strong capture requires—but its value lies in showing that the three conditions are *independently* challenged.

5.2 Independent Lines of Failure

Each condition has been individually challenged from different and largely independent directions.

Representability is challenged from the philosophy of mind. If phenomenal truths are essentially perspectival or first-personal, they may resist the kind of third-person formal encoding that algorithmic theories require Nagel (1974); McGinn (1989). In particular, if phenomenal truths include irreducibly indexical or perspectival elements, their totality may fail to admit a unified formal representation. It has not been established that $\mathrm{Truth}(C) \subseteq \mathcal{R}$.

Algorithmic realizability is challenged from computability theory and cognitive science. The Church–Turing thesis is not formally provable Cleland (1993); the algorithmic status of human cognition is underdetermined (Section 6); and the inference from local computability to global computability depends on compositional assumptions that may not hold for unified phenomenal fields. Even if each component of consciousness admits individual algorithmic capture, the unified phenomenal field need not be algorithmically capturable as a whole:

$$\big(\forall i,\ \exists T_i \in \mathcal{A} : \mathrm{Capture}_{\mathrm{s}}(T_i, c_i)\big) \not\Rightarrow \exists T \in \mathcal{A} : \mathrm{Capture}_{\mathrm{str}}(T, C)$$

This compositionality failure means that successful algorithmic treatment of individual aspects of consciousness does not by itself imply complete algorithmic capture of the whole. The transition from local computability to global capture remains one of the deepest unresolved issues in the theory of consciousness, connected to both the binding problem and the combination problem Chalmers (2016).

A higher-order interpretation may grant this non-implication while arguing that some components of consciousness are themselves cognitive or representational, and therefore algorithmically tractable. Suppose, schematically, that:

$$\mathrm{Truth}(C) = \mathrm{Truth}_{\mathrm{func}}(C) \cup \mathrm{Truth}_{\mathrm{phen}}(C)$$

Here $\mathrm{Truth}_{\mathrm{func}}(C)$ denotes truths about functional or higher-order organization, and $\mathrm{Truth}_{\mathrm{phen}}(C)$ denotes truths about phenomenal character proper. The decomposition is heuristic: it marks a conceptual distinction, not two independent domains. Functional or higher-order components may be tractable even if phenomenal components are not. The framework is compatible with this possibility, but it does not change the core inference:

$$\text{Truth}_{\text{func}}(C) \in \mathcal{A} \not\Rightarrow \text{Truth}(C) \in \mathcal{A}$$

This implication states that algorithmic capturability of the functional component is insufficient for algorithmic capturability of the full explanandum. At most, it shows that some local subcomponents of the explanandum are algorithmically capturable. It does not by itself establish global strong capture, because explanatory closure still requires explaining why the relevant representational organization is accompanied by experience rather than constituting a merely computational relation among representations. The corresponding philosophical issue—how this relocation of burden should be interpreted within higher-order approaches—is developed further in Section 8.3.

Explanatory closure is challenged from the phenomenology of the explanatory gap. No existing theory—including GWT, IIT, and RPT—has been shown to explain why its identified mechanisms are accompanied by experience rather than occurring without phenomenal character (Section 3.5). The gap is not between data and theory but between mechanism and experience.

These three lines of failure are *independent*. The representability challenge arises from the structure of phenomenal facts. The realizability challenge arises from formal constraints on computation. The explanatory closure challenge arises from the nature of the explanatory demand itself. They do not reduce to a single underlying difficulty, nor does progress on one automatically yield progress on the others.

5.3 A2 as the Most Parsimonious Interpretation

Many readers will take algorithmic capturability as the default position, given the broad success of computational methods across the sciences. The burden of the present argument is to show that the evidence specific to consciousness provides reason to depart from this default.

The relationship between A2′ and A2 is not symmetrical. If A2 is true—if complete strong algorithmic capture is impossible in principle—then A2′ follows immediately: a principled impossibility entails a current-state failure. If A2 is false—if complete strong capture is achievable in principle—then the persistent failure to satisfy *any* of the three conditions, despite sustained effort across multiple disciplines, requires an independent explanation for each gap.

The hypothesis that A2 is true provides a single, unified explanation for why all three conditions fail simultaneously: they fail because Truth(C) is not the kind of object that admits complete algorithmic capture. The hypothesis that A2 is false must treat each failure as a separate contingent gap, with no common explanation for their co-occurrence.

On standard principles of inference to the best explanation, a hypothesis that unifies independently observed phenomena provides stronger support than one that treats each as coincidental. The claim advanced here is not that A2 is proved, but that it is the more parsimonious interpretation of the evidence currently available—and that the framework makes this assessment precise by specifying exactly what the evidence consists of and exactly what would be required to overturn it.

5.4 The Profile of the Field

The pattern of progress within consciousness science is itself evidentially relevant. A recent comprehensive review describes the current state of the field as an "uneasy stasis," noting over two hundred distinct theoretical approaches exhibiting wide divergence in metaphysical assumptions and explanatory strategies, with most empirical research designed to support rather than test or compare theories Cleeremans et al. (2025). The taxonomy of Kuhn (2024) identifies approximately 225 theories organized into ten categories spanning quantum mechanics to cosmic consciousness. The most rigorous adversarial test to date—the COGITATE collaboration—found that its results challenged key tenets of both leading theories rather than favoring either Cogitate Consortium et al. (2025).

This is not the profile of a field that is merely young. It is the profile of a field that is empirically productive but *explanatorily fragmented*. If the problem were primarily one of insufficient data, one would expect theoretical convergence as data accumulates. Instead, theoretical divergence has increased alongside empirical progress. This asymmetry—advancing weak and structural capture alongside stalled explanatory closure—is precisely what the present framework predicts under A2, and is more naturally explained by a structural barrier at the level of strong capture than by a temporary data deficit.

5.5 Acknowledging the Pessimistic Meta-Induction

This argument has the structure of a pessimistic meta-induction, and the limitations of such arguments must be acknowledged. Analogous arguments have historically proven unreliable: the persistent failure to reduce biological phenomena to physics was once taken as evidence of vitalism; the persistent failure to observe atoms was taken as evidence of their non-existence. Three decades of modern consciousness science is not long for a problem of this magnitude.

The present argument differs from generic pessimistic meta-inductions in one structural respect: the framework specifies *exactly* what would need to change. It is not a vague claim that "we haven't explained consciousness and probably never will." It identifies three specific, independently assessable conditions and claims that all three must be satisfied simultaneously. This makes the argument falsifiable in a way that generic pessimism is not: a future theory that demonstrably achieves representability, algorithmic realizability, and explanatory closure would refute both A2′ and A2. The framework itself provides the criteria for its own overthrow.

Nevertheless, it remains possible that the right conceptual tools have simply not yet been developed. The balance of evidence, as currently assessed against the specific conditions the framework identifies, favors A2 over its negation. This is a defeasible judgment, not a proof.

5.6 If A2 Holds: The Unconditional Consequence

If the principled thesis A2 is correct—if $\mathrm{Truth}(C) \notin \mathcal{A}$, meaning phenomenal consciousness is not completely algorithmically capturable—then the conditional nature of the main theorem drops away. A2 entails A2′, and the result follows unconditionally (given A1 and A3):

Corollary 3. *If* $\mathrm{Truth}(C) \notin \mathcal{A}$, $\mathcal{E} \subseteq \mathcal{A}$, *and complete explanation requires* $\mathrm{Capture}_{\mathrm{str}}(T, C)$, *then no admissible explanation completely explains phenomenal consciousness.*

Under A1, the space of admissible explanation is algorithmic. Under A2, phenomenal consciousness is not algorithmically capturable. The result is then a mismatch between the explanatory tools available and the nature of the explanandum: the formal framework within which human explanation operates—the algorithmic—cannot completely capture the phenomenon it is directed at.

This is not a claim about the limits of intelligence, creativity, or scientific ingenuity. It is a claim about a structural mismatch between a class of formal methods and a class of phenomena. Weak and structural capture remain achievable; empirical progress continues; partial understanding deepens. What is foreclosed, if A2 holds, is the specific achievement of complete strong capture: the conjunction of explanatory closure and truth-completeness within the algorithmic framework.

This question is not settled here. What the framework provides is (a) a precise characterization of what A2 requires and what it excludes, (b) a decomposition of the conditions that would refute it, and (c) an argument that the current evidence, evaluated against those conditions, favors A2 as the more parsimonious interpretation.

5.7 What Would Refute A2′ and A2

To refute A2′, one must demonstrate all of the following:

1. That $\mathrm{Truth}(C)$ is fully representable within a formal system,
2. That the mapping between phenomenal and physical descriptions is algorithmically derivable,
3. That this mapping yields genuine explanatory closure rather than redescription,
4. That structural reproduction suffices for phenomenal instantiation.

At present, none of these has been conclusively established. Future work that satisfies all four would refute both A2′ and A2 and render the main result inapplicable—but it would do so by achieving precisely what the present paper identifies as missing. The framework is designed to be overturned by the right kind of success.

6 The Algorithmic Status of Explanation

A central question underlying the framework is whether human reasoning itself is algorithmic. The Lucas–Penrose argument (1961; 1989; 1994), based on Gödel's incompleteness theorem, claims that human reasoners can transcend any formal system, but this has been widely criticized for assuming perfect consistency and reliability LaForte et al. (1998); Kerber (2005); Benacerraf (1967). The Computational

Theory of Mind Rescorla (2025); Putnam (1960); Fodor (1975) takes the opposing view, supported by the Church–Turing thesis Turing (1936); Copeland (2024), but the thesis is not formally provable Cleland (1993). Searle's Chinese Room argument (1980; 1992) challenges not computability per se but the sufficiency of computation for genuine understanding. Hypercomputation proposals Siegelmann (1999) suggest the brain might exceed the Turing limit, but rely on physically unrealizable idealizations Davis (2004); Copeland (2002).

The algorithmic status of human cognition is therefore currently underdetermined: neither $M \in \mathcal{A}$ nor $M \notin \mathcal{A}$ is provable. This has two consequences. First, it prevents any unconditional claim that explanation must be algorithmic, reinforcing the conditional nature of A1. Second, even if cognition is non-algorithmic, explanations as communicable and formalizable objects may still be constrained to algorithmic forms—so A1 remains defensible even under cognitive non-computability.

A stronger position bears on this question. Penrose (1994); Penrose and Hameroff (2011) argue that consciousness depends on non-computable physical processes (quantum coherence in microtubules). If such views are correct, then no classical computational system can instantiate consciousness: $C \notin \mathcal{A} \Rightarrow \forall S \in \mathcal{A},\ \neg\,\mathrm{Inst}(S, C)$. The present paper does not assume this thesis, but it represents a live possibility that would independently support A2.

7 Realization, Simulation, and Attribution

The question of whether consciousness can be built is distinct from whether it can be explained. Even if complete strong explanation fails, it may still be possible to predict, simulate, reproduce, or instantiate consciousness.

7.1 Four Levels of Realization

Four increasingly strong notions should be distinguished.

Prediction ($\mathrm{Pred}(S, C)$): a system reproduces the behavior or outward signatures associated with consciousness. *Simulation* ($\mathrm{Sim}(S, C)$): a system models the dynamics or organization of a conscious system. *Structural reproduction* ($\mathrm{Struct}(S, C)$): a system realizes the relevant organization. *Instantiation* ($\mathrm{Inst}(S, C)$): phenomenal consciousness is genuinely present.

These form a conceptual hierarchy:

$$\mathrm{Inst}(S, C) \Rightarrow \mathrm{Struct}(S, C) \Rightarrow \mathrm{Sim}(S, C) \Rightarrow \mathrm{Pred}(S, C)$$

but not conversely. The most contentious link is the first: Inst $\Rightarrow$ Struct presupposes that consciousness has an identifiable structural basis whenever it is present. A strongly emergentist or primitivist view might deny this—but such a view would undermine the prospect of any structural or mechanistic theory, which would strengthen rather than weaken the paper's central claim. In particular, if consciousness lacks identifiable structural correlates, representability ($\mathrm{Truth}(C) \subseteq \mathcal{R}$) becomes harder to establish, reinforcing rather than undermining the triple constraint.

The critical non-implications are:

$$\mathrm{Pred}(S, C) \nRightarrow \mathrm{Inst}(S, C) \qquad \text{and} \qquad \mathrm{Struct}(S, C) \nRightarrow \mathrm{Inst}(S, C)$$

The first is widely accepted: behavioral mimicry is insufficient for consciousness Searle (1980); Bengio and Elmoznino (2025); Schwitzgebel (2025). As Searle observed, a simulation of digestion does not digest; likewise, a simulation of consciousness need not be conscious. The second is precisely one of the central disputed claims in the literature. Computational functionalism affirms the inference from structure to instantiation Chalmers (1996); biological naturalism denies it Searle (1992); Seth (2025). IIT explicitly rejects the inference from functional equivalence to conscious equivalence, holding that two functionally identical systems may differ in Φ and therefore in consciousness Tononi et al. (2016); Albantakis et al. (2023).

Whether the inference from structural reproduction to instantiation holds remains underdetermined Birch (2025); Butlin et al. (2023, 2025). Formally:

$$\begin{aligned} &\neg\,\mathrm{Provable}\big(\mathrm{Struct}(S, C) \Rightarrow \mathrm{Inst}(S, C)\big) \\ \wedge\ &\neg\,\mathrm{Provable}\big(\mathrm{Struct}(S, C) \nRightarrow \mathrm{Inst}(S, C)\big) \end{aligned}$$

Recent empirical work reinforces this concern. Larkum et al. (2025) identify cases in which ongoing neural dynamics appear to diverge from the underlying computational organization, suggesting a possible disconnect between computation and phenomenology. Their results raise the possibility that even structurally faithful computational models may fail to capture the processes relevant to conscious experience. While these findings do not decisively refute computational theories, they reinforce the need to distinguish carefully between simulation, structural reproduction, and instantiation.

Recent work also proposes shifting from behavioral tests to theoretically grounded indicators of consciousness—evaluating systems against properties derived from leading neuroscientific theories such as global workspace dynamics, recurrent processing, and integrated information Butlin et al. (2025). These indicator-based approaches aim to bridge prediction and structural reproduction, but they do not eliminate the distinction between structural reproduction and instantiation.

7.2 Three Levels of Structural Reproduction

The literature suggests that objections to artificial consciousness operate at multiple levels: some concern algorithms, others concern architecture, and still others concern the physical substrate itself Multiple Authors (2025). This motivates a finer decomposition of structural reproduction:

$$\mathrm{Struct}(S, C) \in \{\mathrm{Struct}_{\mathrm{func}}, \mathrm{Struct}_{\mathrm{causal}}, \mathrm{Struct}_{\mathrm{sub}}\}$$

where the relevant notion of structure may be functional, causal, or substrate-bound. The functionalist position tends to privilege $\mathrm{Struct}_{\mathrm{func}}(S, C)$—the view that reproducing the right functional organization suffices for consciousness Chalmers (1996). Biological naturalism and IIT-style objections often require stronger conditions involving causal or substrate-level equivalence Searle (1992); Tononi et al. (2016). This decomposition makes precise the level at which different positions disagree about the sufficiency of structural reproduction for instantiation.

7.3 Attribution Without Instantiation

The non-implication between prediction and instantiation is not merely theoretical. Large-scale studies show that human observers systematically attribute phenomenal consciousness to systems exhibiting only prediction-level behavioral success. Colombatto and Fleming (2024) report that approximately two-thirds of respondents attributed at least some degree of phenomenal consciousness to ChatGPT, with attributions scaling with usage frequency and perceived phenomenal qualities rather than perceived intelligence. Kang et al. (2025) show that metacognitive self-reference and apparent emotions increase perceived consciousness, while factual competence reduces it.

These attributions are driven by anthropomorphic tendencies rather than structural understanding Epley et al. (2007); Nass and Moon (2000); Pataranutaporn et al. (2025); Seth (2025). Consciousness attribution is better understood as a property of the observer's cognitive processes than as evidence of phenomenal states in the attributed system. As artificial systems become increasingly effective at simulating the outward signatures of conscious agents, the distinction between prediction, structural reproduction, and instantiation may become harder to detect—and success at prediction may be misinterpreted as evidence for instantiation or even complete explanation.

The framework separates the question of explanation from the psychology of attribution by defining complete explanation in terms of strong capture rather than behavioral adequacy.

8 Discussion

The preceding analysis identifies multiple independent sources of limitation. These are not merely theoretical; they also structure the practical problem of how human observers interpret artificial systems, often conflating predictive adequacy with phenomenal presence. These limitations span at least eight distinct dimensions:

- *Computational limitation*: the possibility that complete explanation exceeds algorithmic derivation,
- *Representational limitation*: the possibility that phenomenal truths are not fully formalizable,
- *Conceptual limitation*: the explanatory gap between structure and experience,
- *Cognitive limitation*: the underdetermined algorithmic status of human reasoning,
- *Explanatory limitation*: the gap between weak or structural success and strong capture,

- *Instantiational limitation*: the gap between simulation, structural reproduction, and phenomenal presence,
- *Attributional limitation*: the tendency of observers to infer consciousness from behavior in ways that outrun the evidential basis for instantiation,
- *Scope limitation*: the paper's results are conditional on phenomenal realism; under illusionism, the target of explanation dissolves rather than being shown to resist explanation.

The multi-dimensional character of this list is itself significant. The obstacle to complete explanation is not a single bottleneck but a convergence of independent constraints operating at different levels of analysis.

8.1 Type-A Physicalism and the Dissolution Strategy

The most radical objection holds that there is no genuine hard problem—that once the physical and functional facts are fully specified, there is nothing further to explain. This position, most extensively developed by Dennett (1991, 2017), holds that the appearance of an explanatory gap is a cognitive artifact generated by confused philosophical intuitions. On this view, A3 is false and the main result is inapplicable.

The framework treats this as a scope condition. The analysis is addressed to those who hold that phenomenal consciousness poses a genuine explanatory problem—the position shared by Nagel (1974), Levine (1983), Chalmers (1996), and the majority of philosophers of mind. Dennett's position does not offer an alternative theory of complete strong capture; it dissolves the demand. Within the present framework, his view amounts to denying A3 rather than satisfying the conditions for strong capture. The two positions are orthogonal.

8.2 Type-B Physicalism and the Explanatory Gap

Type-B physicalism accepts the explanatory gap as genuine but interprets it as a limitation of human cognition rather than evidence of a metaphysical distinction Levine (1983). The present framework is compatible with this interpretation. If explanatory closure is systematically unavailable to cognitive agents—whether for metaphysical or cognitive reasons—then complete strong capture remains unavailable in practice. The limitation applies at the level of explanation regardless of its ultimate metaphysical source. This is significant: even a type-B physicalist who believes consciousness *is* physical must concede that no current theory achieves the explanatory closure required for strong capture.

8.3 Higher-Order Thought and the Location of the Gap

An important higher-order position holds that a mental state is phenomenally conscious only if it stands in a further cognitive relation, typically higher-order representation or thought Rosenthal (2005); Brown et al. (2019). This position admits two readings.

On an *epistemic* reading, higher-order cognition is required to access, report, or introspect phenomenal states, but not to constitute their existence. Under that reading, the issue concerns epistemic access rather than metaphysics. The framework is largely unchanged: $\mathrm{Truth}(C)$ is defined extensionally as whatever the phenomenal facts are, independent of how they are detected or reported.

On a *constitutive* reading, higher-order cognition is taken to be partly constitutive of phenomenal consciousness itself. This is the stronger challenge, because it suggests that algorithmically tractable cognitive structure may account for substantial portions of the explanandum. However, the effect is primarily to relocate the explanatory burden. As noted in Section 3.5, even if one grants higher-order architecture, the key question remains why that architecture is accompanied by experience rather than amounting to a purely computational relation among representations. The explanatory gap shifts level rather than disappearing.

This also clarifies the role of $\mathrm{Func}(C)$. If higher-order cognition is a necessary condition on phenomenal occurrence, then functional structure may be necessary for C while still not being sufficient for explanatory closure. Formally, one may grant necessity without granting identity:

$$\mathrm{Func}(C) \text{ necessary for } C \not\Rightarrow \mathrm{Func}(C) = \mathrm{Truth}(C)$$

The expression says that even if functional organization is required whenever phenomenal consciousness occurs, it does not follow that functional truths exhaust all truths about phenomenal consciousness. Accordingly, the framework's central strategy is meta-theoretic: wherever the gap is located—between physics and phenomenology, between cognition and phenomenology, or between higher-order representation and phenomenology—the same triple constraint recurs. A constitutive higher-order position need not itself be type-A in Dennett's sense; however, insofar as it is presented as delivering complete strong capture with no residual phenomenal explanandum, it becomes dialectically equivalent to a type-A dissolution strategy (Section 8.1), which this paper treats as a scope position rather than a refutation.

8.4 Intertheoretic Reduction

The philosophy of science offers models of intertheoretic reduction: Nagel's bridge-law model (1961), Schaffner's generalized reduction-replacement model (1967), and Kim's supervenience arguments (2005). Consciousness is widely regarded as a hard case for these models precisely because the relevant bridge principles are unavailable. Levine's point (1983) is that psychophysical bridge laws, even if true, lack the explanatory force that bridge laws have in other domains: knowing that C-fiber firing is identical to pain does not explain *why* C-fiber firing feels like pain. Standard models of reduction presuppose that bridge principles, once stated, render the reduction explanatory. In the case of consciousness, this presupposition is what is contested.

The present framework translates this concern precisely: existing theories achieve structural capture but not explanatory closure. If a future reduction provided genuinely explanatory bridge principles, it would satisfy the conditions for strong capture. The triple constraint identifies what such a reduction would need.

8.5 Mathematical Consciousness Science

Recent formal approaches represent the most promising potential challenge to A2′. Kleiner and Hoel (2021) have developed mathematical frameworks for consciousness science, and Tsuchiya et al. (2016) apply category-theoretic tools to the structure of conscious experience. These approaches aim to provide formal, mathematically rigorous accounts of phenomenal structure.

Evaluated against the triple constraint: current mathematical consciousness science has made progress on representability (formalizing aspects of phenomenal structure) and potentially on algorithmic realizability (working within computable mathematical frameworks). Whether these approaches achieve explanatory closure—whether they explain *why* the formalized structures are accompanied by experience—remains open. Mathematical precision in the description of phenomenal structure is an advance in structural capture, but it does not by itself close the explanatory gap. These approaches therefore represent the frontier at which A2′ is most likely to be challenged.

8.6 The Meta-Problem

Chalmers (2018) introduces the meta-problem: explaining why we think there is a hard problem. A theory might explain why cognitive systems generate problem-intuitions without explaining phenomenal consciousness itself. In the terms adopted here, the meta-problem targets truths about cognitive and functional processing:

$$\mathrm{Meta}(C) \subseteq \mathrm{Func}(C)$$

A solution to the meta-problem would explain why systems like us produce reports and intuitions about consciousness. But it would not thereby explain why phenomenal consciousness exists. In formal terms:

$$\mathrm{ExplClosure}(T, \mathrm{Meta}(C)) \not\Rightarrow \mathrm{ExplClosure}(T, C)$$

Explaining why we *think* there is an explanatory gap is not the same as closing the gap. A theory that accounts for problem-intuitions achieves functional or structural capture of the meta-problem, but this does not constitute strong capture of phenomenal consciousness itself. The meta-problem shows that certain aspects of the *discourse* about consciousness are algorithmically tractable, while leaving the target of the present paper—the explanation of phenomenal consciousness—untouched.

8.7 Illusionism

Illusionism Frankish (2016) maintains that phenomenal consciousness as defined here does not exist. If correct, C does not exist and the framework loses its target. The paper treats phenomenal realism as a scope condition. Two observations: first, illusionism does not eliminate the explanatory burden but relocates it—the "illusion problem" may be at least as hard as the original Kammerer (2016). Second, explaining why we *judge* ourselves to be phenomenally conscious does not establish that phenomenal consciousness is illusory; it establishes only that the judgment is functionally explicable.

8.8 Scope, Modesty, and the Moving Target

Scientific explanations are typically partial and revisable. Requiring full coverage of Truth(C) may seem unrealistically strong Wu and Bhatt (2025); Seth and Bayne (2022). But the levels of capture are distinguished precisely to preserve the legitimacy of partial explanations. The limitation result targets a specific ambition: complete explanation that closes the explanatory gap. A related concern is that the target may shift as research progresses Chalmers (2018); Wagner-Altendorf (2024). The framework requires only that a complete explanation range over the full truth-set, however ultimately characterized.

8.9 Limitations

The main result is conditional on A1–A3. The argument that A2 is the most parsimonious interpretation is inductive and defeasible. The framework does not refute reductionist positions but clarifies what they must establish. The argument depends on phenomenal realism. The paper does not resolve whether artificial systems can instantiate consciousness. The inductive argument may prove analogous to pessimistic meta-inductions that were eventually overturned—though the framework's specificity (identifying exact refutation conditions) distinguishes it from generic pessimism.

9 Conclusion

This paper has developed a framework of capture levels for evaluating theories of consciousness and identified three independently motivated constraints—representability, algorithmic realizability, and explanatory closure—whose simultaneous satisfaction is required for complete strong algorithmic capture. The persistent and independent failure of all three, evaluated against the framework's specific criteria, supports the following assessment: the principled thesis A2—that phenomenal consciousness is not completely algorithmically capturable—is the most parsimonious interpretation of the available evidence.

This is a defeasible claim. The framework specifies exactly what would need to be demonstrated to overturn it: representability, algorithmic realizability, and explanatory closure, achieved simultaneously. A future theory that satisfies all three conditions would refute both the operational premise A2′ and the principled thesis A2. But the current state of the field—empirically productive at the level of weak and structural capture, explanatorily stalled at the level of strong capture, and theoretically fragmenting rather than converging—is better explained by a structural limitation than by three independent contingent gaps.

If A2 holds, the conditional limitation becomes unconditional: the algorithmic framework within which human explanation operates cannot completely capture phenomenal consciousness. This is not a counsel of despair. Weak and structural capture remain achievable, empirical progress continues, and the framework itself provides a precise map of what remains missing. But it does suggest that the hard problem of consciousness may be hard in a deeper sense than is often acknowledged—not merely unsolved, but resistant to the class of solutions that the explanatory tools available to us can provide.

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